Day 07: IP Configuration

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Cybersecurity Learning Log

Part 1: Theory Questions and Answers

1. What is a network address?
A network address identifies a particular network. It is used to distinguish the network from the individual devices connected to it.

2. What is a broadcast address?
A broadcast address is used to send a packet to all hosts on a particular IPv4 network.

3. What is a usable host address?
A usable host address is an IP address that can normally be assigned to a device on a network, rather than being reserved as the network or broadcast address.

4. What is the first usable IP address?
It is normally the address immediately after the network address in a subnet.

5. What is the last usable IP address?
It is normally the address immediately before the broadcast address in a subnet.

6. What does /24 mean?
The /24 prefix means that the first 24 bits of the IPv4 address identify the network, leaving 8 bits for host addresses.

7. What subnet mask corresponds to /24?
The subnet mask is 255.255.255.0.

8. What is a default gateway?
A default gateway is usually a router that forwards traffic from a local network to other networks when there is no more specific route.

9. Why does a router interface need an IP address?
An IP address allows the interface to communicate at Layer 3 and act as a reachable endpoint on its connected IP network.

10. What is a static IP configuration?
A static configuration uses manually assigned network settings, such as an IP address, subnet mask, gateway, and DNS server, rather than receiving them automatically through DHCP.

11. What does show ip interface brief display?
On Cisco IOS devices, it provides a quick summary of interfaces, their IP addresses, and their interface and line-protocol status.

12. What does no shutdown do conceptually on a Cisco interface?
It removes the administrative shutdown state and enables the interface, provided the hardware and connection are working.

13. What does interface status up/up generally indicate?
It generally means the interface is operational and its line protocol is also operational.

14. Why should network configuration always be verified?
Verification helps identify incorrect addresses, subnet masks, gateways, DNS settings, or interface states before they cause connectivity or security problems.

15. Why is IPv4 configuration important in cybersecurity?
Correct IPv4 configuration supports network monitoring, traffic analysis, access control, troubleshooting, and the identification of unexpected or suspicious network activity.

Part 2: IPv4 Exercise (192.168.100.0/24)

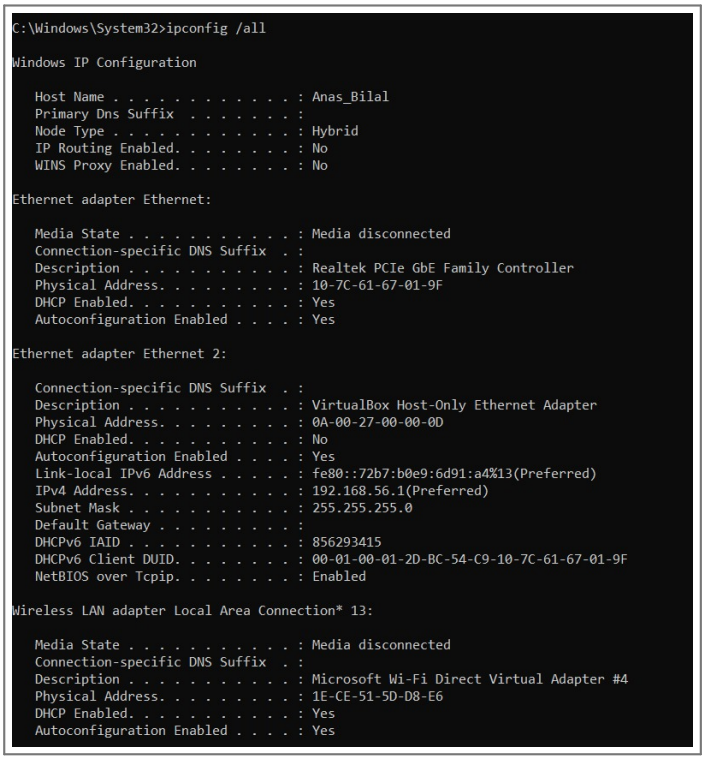
Item	Address
Network Address	192.168.100.0
First Usable IP	192.168.100.1
Last Usable IP	192.168.100.254
Broadcast Address	192.168.100.255

Part 3: Practical (ipconfig /all)

Item	Value	Source
IPv4 Address	192.168.1.38	Figure 2
Subnet Mask	255.255.255.0	Figure 2
Default Gateway	192.168.1.1	Figure 2
DNS Server	8.8.8.8 (primary), 8.8.4.4 (secondary)	Figure 2

Part 4: Screenshots

Figure 1: ipconfig /all - Host Details and Ethernet Adapters



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C:\Windows\System32>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Anas_Bilal
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : 10-7C-61-67-01-9F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

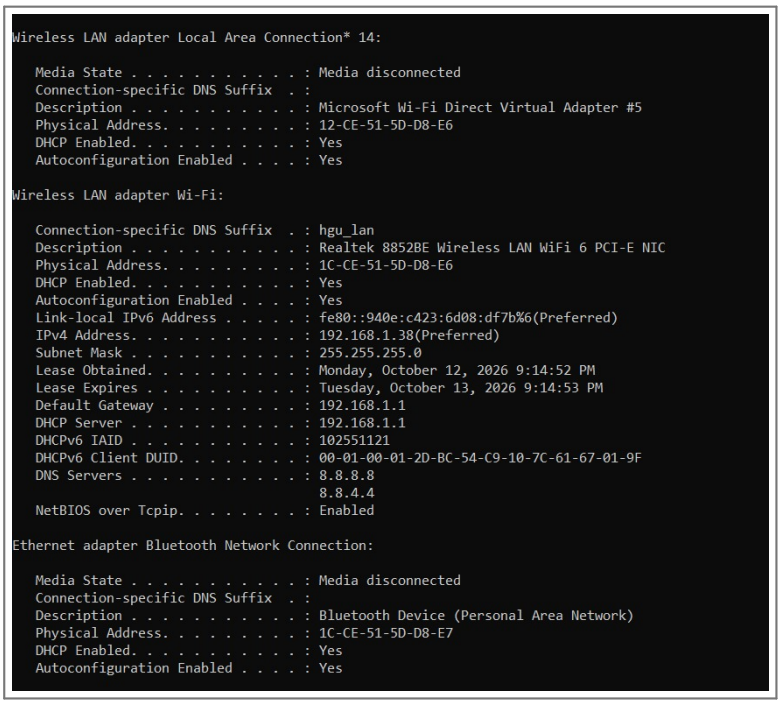
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-0D
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::72b7:b0e9:6d91:a4%13(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 856293415
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-BC-54-C9-10-7C-61-67-01-9F
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 13:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #4
   Physical Address. . . . . . . . . : 1E-CE-51-5D-D8-E6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

Shows the host details and the Ethernet adapters. Ethernet is disconnected, and Ethernet 2 is a VirtualBox Host-Only adapter with 192.168.56.1.

Figure 2: ipconfig /all - Active Wi-Fi Adapter



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Wireless LAN adapter Local Area Connection* 14:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #5
   Physical Address. . . . . . . . . : 12-CE-51-5D-D8-E6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

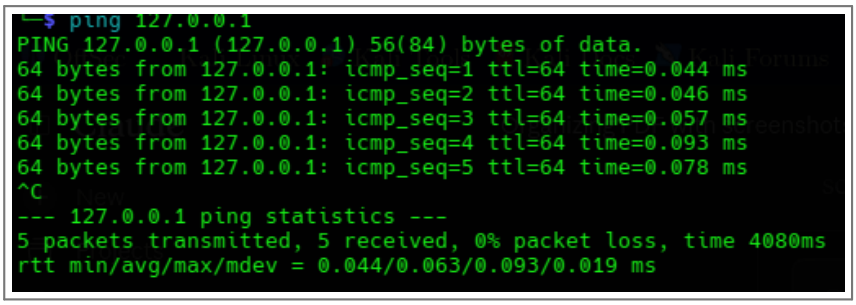
   Connection-specific DNS Suffix  . : hgu_lan
   Description . . . . . . . . . . . : Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
   Physical Address. . . . . . . . . : 1C-CE-51-5D-D8-E6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::940e:c423:6d08:df7b%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.38(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Monday, October 12, 2026 9:14:52 PM
   Lease Expires . . . . . . . . . . : Tuesday, October 13, 2026 9:14:53 PM
   Default Gateway . . . . . . . . . : 192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 102551121
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-BC-54-C9-10-7C-61-67-01-9F
   DNS Servers . . . . . . . . . . . : 8.8.8.8
                                       8.8.4.4
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 1C-CE-51-5D-D8-E7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
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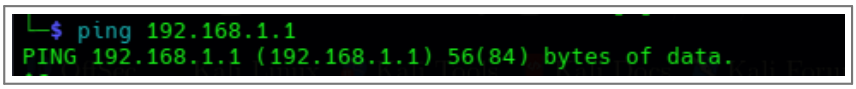
The active Wi-Fi adapter with IPv4 address 192.168.1.38, subnet mask 255.255.255.0, default gateway 192.168.1.1 and DNS servers 8.8.8.8 and 8.8.4.4.

Figure 3: Loopback Test (ping 127.0.0.1)



5 packets sent, 5 received, 0% packet loss. The local TCP/IP stack responds.

Figure 4: Ping to Default Gateway (ping 192.168.1.1)



The ping to the default gateway 192.168.1.1 was started. This capture shows the command and its first line, with no replies yet.